Example 5.32

Calculate the quartiles Q_1 and Q_3 for wages of the labours given below

Wages (Rs.)	30-32	32-34	34-36	36-38	38-40	40-42	42-44
Labourers	12	18	16	14	12	8	6

Solution:

x	f	cf
30 – 32	12	12
32 – 34	18	30
34 – 36	16	46
36 – 38	14	60
38 – 40	12	72
40 – 42	8	80
42 – 44	6	86
	86	

$$\frac{N}{4} = \frac{86}{4} = 21.5$$

lies in the group 32 – 34

$$Q_1 = l_1 + \frac{\frac{N}{4} - m_1}{f_1} \times c_1$$

$$= 32 + \frac{21.5 - 12}{18} \times 2$$

$$= 32 + \frac{19}{18} = 32 + 1.06$$

$$= \text{Rs. } 33.06$$

$$\frac{3N}{4} = \frac{3 \times 86}{4} = 64.5$$

$\therefore Q_3$ lies in the group 38 – 40

$$Q_3 = l_3 + \frac{3\left(\frac{N}{4}\right) - m_3}{f_3} \times C_3$$

$$= 38 + \frac{64.5 - 60}{12} \times 2$$

$$= 38 + \frac{4.5}{12} \times 2$$

$$= 38 + 0.75 = \text{Rs. } 38.75$$

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